

Phần 7 - Documenting Software Architecture

Mục lục

7.1 Tổng quan Phần 7: Documenting Software Architecture	3
Nếu bạn đến từ Data Science	3
Vì sao quan trọng	3
1. Bus factor	3
2. Onboarding	3
3. Architecture evolution	4
Phần 7 sẽ dạy	4
7.2: Module Views	4
7.3: Component-and-Connector Views	4
7.4: Allocation Views	4
Framework chính: 3 views (Bass-Clements-Kazman)	4
Diagram standards	5
C4 Model (Simon Brown)	5
4+1 View (Philippe Kruchten, 1995)	5
UML	5
Mermaid / PlantUML	5
ADR, Architecture Decision Record	5
Practices documentation	6
Living documentation	6
Audience-aware	6
Minimum viable doc	6
Cách đọc phần	6
7.2 Module Views	8
Nếu bạn đến từ Data Science	8
Module là gì trong context view này	8
4 subtypes của Module View	8
Subtype 1: Decomposition View	8
Subtype 2: Dependency View	9
Subtype 3: Generalization View	9
Subtype 4: Uses View	9
C4 Model deep dive	10
Level 1: System Context	10
Level 2: Container	10
Level 3: Component	10
Level 4: Code (optional)	10
Tools cho C4	10

ADR, Architecture Decision Record	14
ADR best practices	15
Tools	15
Practical: Documenting một service	15
Anti-patterns	15
Anti-pattern 1: Big upfront design	15
Anti-pattern 2: Diagram-only doc	15
Anti-pattern 3: No audience awareness	16
Anti-pattern 4: Doc rot	16
Tóm tắt	16
7.3 Component-and-Connector Views	17
Nếu bạn đến từ Data Science	17
Module vs C&C	17
Subtypes của C&C	17
Subtype 1: Process View	17
Subtype 2: Concurrency View	17
Subtype 3: Communication View	17
Subtype 4: Deployment-time	19
Khi nào dùng C&C View	19
Diagram types	19
Sequence Diagram (UML)	19
Activity Diagram (UML)	19
Communication Diagram (UML)	19
Runtime Topology Diagram	20
Connector types	20
Documenting common runtime patterns	20
Pattern 1: Request flow	20
Pattern 2: Failover	20
Pattern 3: Saga (distributed transaction)	20
Pattern 4: Async event flow	20
Sai lầm thường gặp	20
Sai lầm 1: Vẽ mọi sequence	20
Sai lầm 2: Quên label connector	22
Sai lầm 3: Mixed level of abstraction	22
Sai lầm 4: No legend	22
Sai lầm 5: Outdated runtime	22
Tóm tắt	22
7.4 Allocation Views	23
Nếu bạn đến từ Data Science	23
3 subtypes	23
Subtype 1: Deployment View	23
Subtype 2: Implementation View	24
Subtype 3: Work Assignment View	24
Cloud topology patterns	25
Pattern 1: Single region, multi-AZ	25
Pattern 2: Multi-region active-passive	25
Pattern 3: Multi-region active-active	25

Pattern 4: Multi-cloud	25
Pattern 5: Edge + Cloud	25
Container orchestration	26
Kubernetes (K8s)	26
Managed services	26
Serverless	26
Conway's Law	26
Examples	27
Implications	27
Visualization	27
Documenting cloud cost	27
Sai lầm thường gặp	27
Sai lầm 1: Outdated deployment diagram	27
Sai lầm 2: No cost annotation	28
Sai lầm 3: Quên failure scenario	28
Sai lầm 4: Work Assignment không up-to-date	28
Sai lầm 5: Quên Conway's law	28
Tóm tắt	28
Tổng kết Phần 7	28

7.1 Tổng quan Phần 7: Documenting Software Architecture

Tóm tắt một dòng: Kiến trúc không được tài liệu hoá là kiến trúc chết - tồn tại trong đầu 1-2 người, evolve không kiểm soát, mất khi người đó rời team. Phần này dạy framework chuẩn để document - 3 views (Module/C&C/Allocation), ADR, và diagram standards.

Nếu bạn đến từ Data Science

ML/data system càng cần documentation vì failure thường nằm giữa nhiều boundary: data source đổi schema, feature transform khác training, model registry thiếu metadata, serving API dùng model cũ, monitoring không nhận prediction log. Nếu không có architecture view, mỗi incident trở thành một cuộc truy tìm thủ công.

Ít nhất, hãy document một model production bằng 3 view: Module View cho code/pipeline modules, C&C View cho runtime flow của prediction/training, Allocation View cho nơi chạy API, job, registry, warehouse, object storage và monitoring.

Vì sao quan trọng

3 lý do hard-hitting:

1. Bus factor

“Bus factor” = số người trong team rời đi (hypothetically bị bus đụng) để hệ thống không ai biết. Bus factor = 1 nghĩa hệ thống chỉ 1 người hiểu. Đó là disaster waiting to happen.

Document tốt $\square$ bus factor ≥ 3 . Team thay đổi, kiến trúc không bị lost.

2. Onboarding

New engineer join team. Không document $\square$ spend 2-3 tháng “figure out” hệ thống qua đọc code. Có document tốt $\square$ onboard 2 tuần.

Cost difference: ~3 tháng * salary mỗi engineer.

3. Architecture evolution

Quyết định cũ làm vì context cũ. Context thay đổi (vd: business model pivot) □ cần revisit decision. Không document vì sao □ không biết quyết định nào còn valid.

ADR (Architecture Decision Record) chính là giải pháp.

Phần 7 sẽ dạy

Bốn bài:

7.2: Module Views

View về *static structure* của code. Module/package/class hierarchy. Subtypes: decomposition, dependency, generalization, uses. C4 model (Context-Container-Component-Code).

7.3: Component-and-Connector Views

View về *runtime structure*. Component instances, connectors, data flow. Subtypes: process, concurrency, communication, deployment-time.

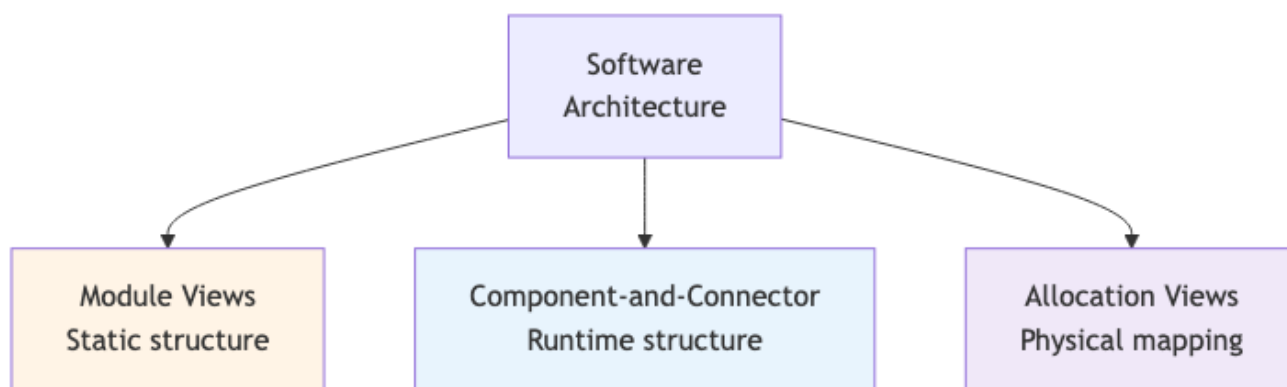
7.4: Allocation Views

View về *physical deployment*. Mapping software □ hardware/network/people. Subtypes: deployment, implementation, work assignment.

Plus: ADR (Architecture Decision Records) lồng vào Bài 7.2.

Framework chính: 3 views (Bass-Clements-Kazman)

SEI Carnegie Mellon trong *Software Architecture in Practice* định nghĩa 3 view fundamental:



Hình 1: Sơ đồ 33

Mỗi view answer câu hỏi khác:

- **Module:** Code organized ra sao? Module nào depend module nào?
- **C&C:** Runtime hệ chạy ra sao? Component nào communicate với component nào, khi nào?

- **Allocation:** Software chạy ở đâu? Mapping vào server/cloud/team?

Documenting đầy đủ ≥ 1 view mỗi loại. Big system có thể có nhiều view mỗi loại.

Diagram standards

Đừng vẽ diagram ad-hoc. Dùng standard để team đọc được:

C4 Model (Simon Brown)

4 mức zoom:

1. **System Context:** hệ trong landscape (user, external system).
2. **Container:** app, database, broker. Mỗi container = deployable unit.
3. **Component:** bên trong container, các component logic.
4. **Code:** class, function (rarely drawn, code is source of truth).

C4 đơn giản, phổ biến, free tool (c4-plantuml, structurizr).

4+1 View (Philippe Kruchten, 1995)

5 view:

- **Logical:** object-oriented design (UML class diagram).
- **Process:** runtime processes (UML sequence/activity).
- **Development:** code organization (UML package/component).
- **Physical:** deployment (UML deployment diagram).
- **Scenarios:** use cases tying everything together.

4+1 chi tiết hơn C4, more formal. Phù hợp enterprise documentation.

UML

Universal Modeling Language. Set of diagram types. Vẫn dùng nhưng heavy. Modern team thường dùng C4 instead.

Mermaid / PlantUML

Tools render diagram from text. Maintainable in git, version-controlled.

graph TD

A[Web App] --> B[API Gateway]

B --> C[Order Service]

Phù hợp markdown documentation.

ADR, Architecture Decision Record

Đã giới thiệu Bài 3.3. Format:

```
# ADR-N: Title
```

```
## Status
```

Proposed | Accepted | Deprecated | Superseded

Context

What's the situation? Why do we need to decide?

Decision

What did we decide?

Consequences

Positive + negative consequences.

Alternatives Considered

What else we considered, why rejected.

ADR file numbered, append-only (don't edit accepted ADRs; superseded by new ADR).

Vì sao ADR works:

- **Lightweight:** < 1 trang. Engineer chịu viết.
- **In git:** version-controlled với code. Diff over time.
- **Searchable:** future engineer find why decision was made.

Tools: adr-tools (CLI), Markdown ADR convention.

Practices documentation**Living documentation**

Doc evolve cùng code. CI verify (vd: link checker, snippet test).

Audience-aware

Different doc cho different audience:

- **Architect / Lead:** full diagram + ADR.
- **Engineer:** code-level doc, API spec.
- **Onboarding:** tutorial, walk-through.
- **Stakeholder / PM:** high-level diagram + narrative.

Minimum viable doc

Đừng over-document. Default:

- 1 System Context diagram.
- 1 Container diagram (C4).
- Module/C&C view cho khu phức tạp.
- ADRs cho important decisions.
- README per repo: setup, build, contribute.

Không cần full UML mọi class.

Cách đọc phần

Tuần tự 7.2 □ 7.3 □ 7.4. Mỗi bài ~3000-4000 chữ.

Sau phần, bạn nên:

- Vẽ được Module View + C&C View + Deployment View cho hệ mình đang làm.
- Viết được 5 ADR cho 5 quyết định gần nhất.
- Setup C4 diagram trong markdown của repo.

Bài tiếp: [Module Views](#).

7.2 Module Views

Tóm tắt một dòng: Module View show static structure của code - module nào tồn tại, depend module nào. Quan trọng cho developer hiểu codebase và cho architect verify modularity (Bài 3.4).

Nếu bạn đến từ Data Science

Module View giúp biến project ML từ một thư mục notebook lộn xộn thành cấu trúc có ý nghĩa. Ví dụ data, features, training, evaluation, registry, serving, monitoring là các module có trách nhiệm khác nhau. Khi nhìn module view, người mới biết code feature nằm đâu, code inference nằm đâu, và boundary nào không nên phá.

Nếu module view không rõ, training và serving thường copy logic của nhau. Đó là nguồn gốc của training-serving skew.

Module là gì trong context view này

Module = implementation unit. Có thể là:

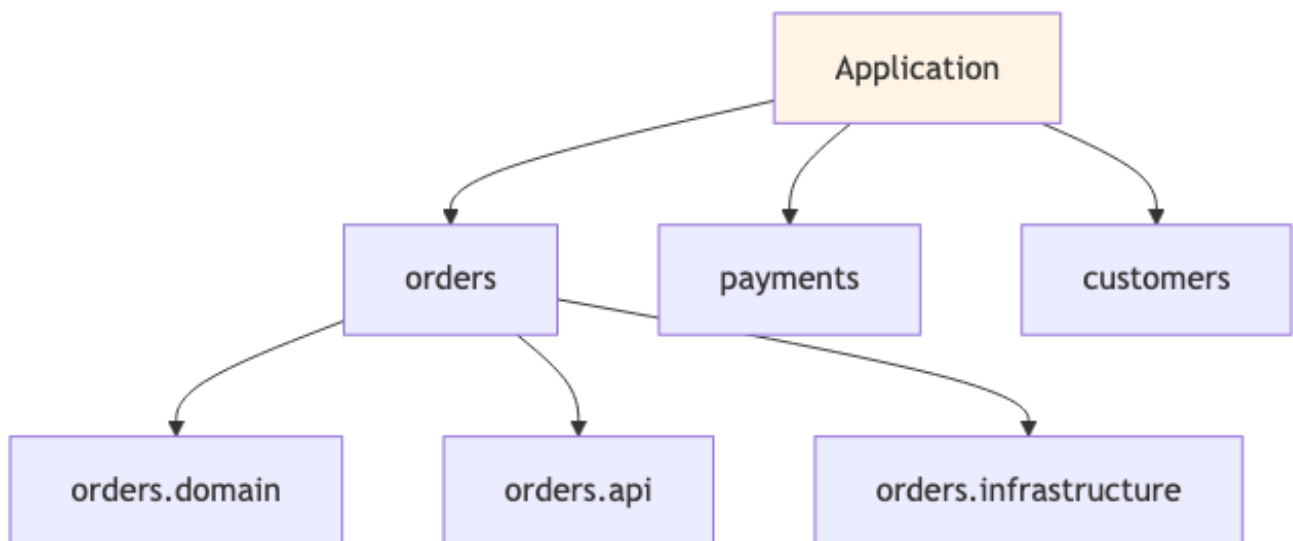
- Package / namespace (Java package, Python module, C# namespace).
- Library / module (npm package, Maven module).
- Class (lớn).
- Subsystem.

Scale: bài này focus mức package/library, không quá fine-grained (không vẽ mọi class).

4 subtypes của Module View

Subtype 1: Decomposition View

Hiển thị hierarchy “module này chứa các sub-module nào”.

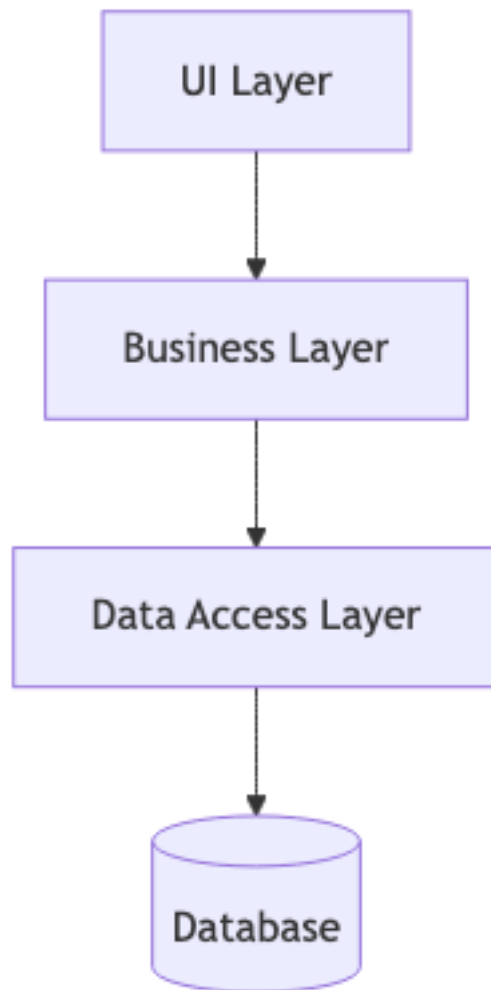


Hình 2: Sơ đồ 34

Purpose: navigation. Developer mới hiểu “code organize ở đâu”.

Subtype 2: Dependency View

Show “module nào depend module nào”.



Hình 3: Sơ đồ 35

Purpose: verify modularity. Architect check: có cycle không? Direction có đúng không?

Tools: madge (JS), pydeps (Python), jdeps (Java) generate auto.

Subtype 3: Generalization View

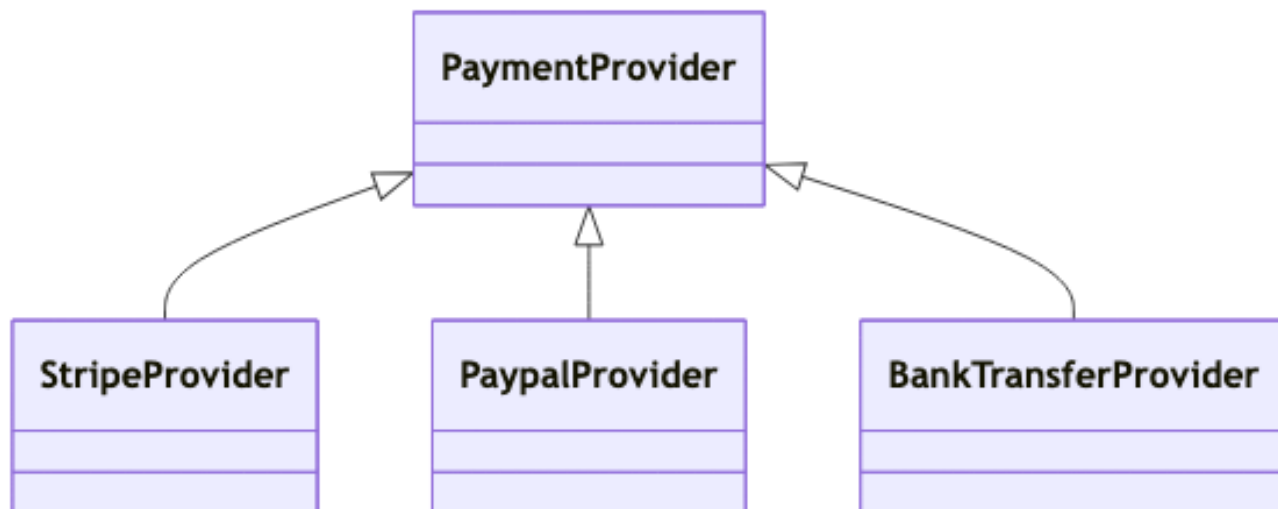
Show inheritance / interface hierarchy.

Purpose: design review. Verify OCP/LSP applied correctly.

Subtype 4: Uses View

Show “module này uses module nào” (richer than dependency). Ai gọi ai ở mức operation.

Purpose: impact analysis. “Nếu sửa module X, module nào có thể bị ảnh hưởng?”.



Hình 4: Sơ đồ 36

C4 Model deep dive

C4 (Simon Brown) là chuẩn modern phổ biến nhất. 4 mức:

Level 1: System Context

Show hệ ở giữa, surrounded by users + external systems.

Audience: stakeholder, PM. Mục đích: “bigger picture”.

Level 2: Container

Zoom vào System box □ show containers. Container = deployable unit (web app, mobile app, database, message broker).

Audience: architect, dev lead. Mục đích: hiểu tech stack + deployment.

Level 3: Component

Zoom vào 1 container □ show components bên trong.

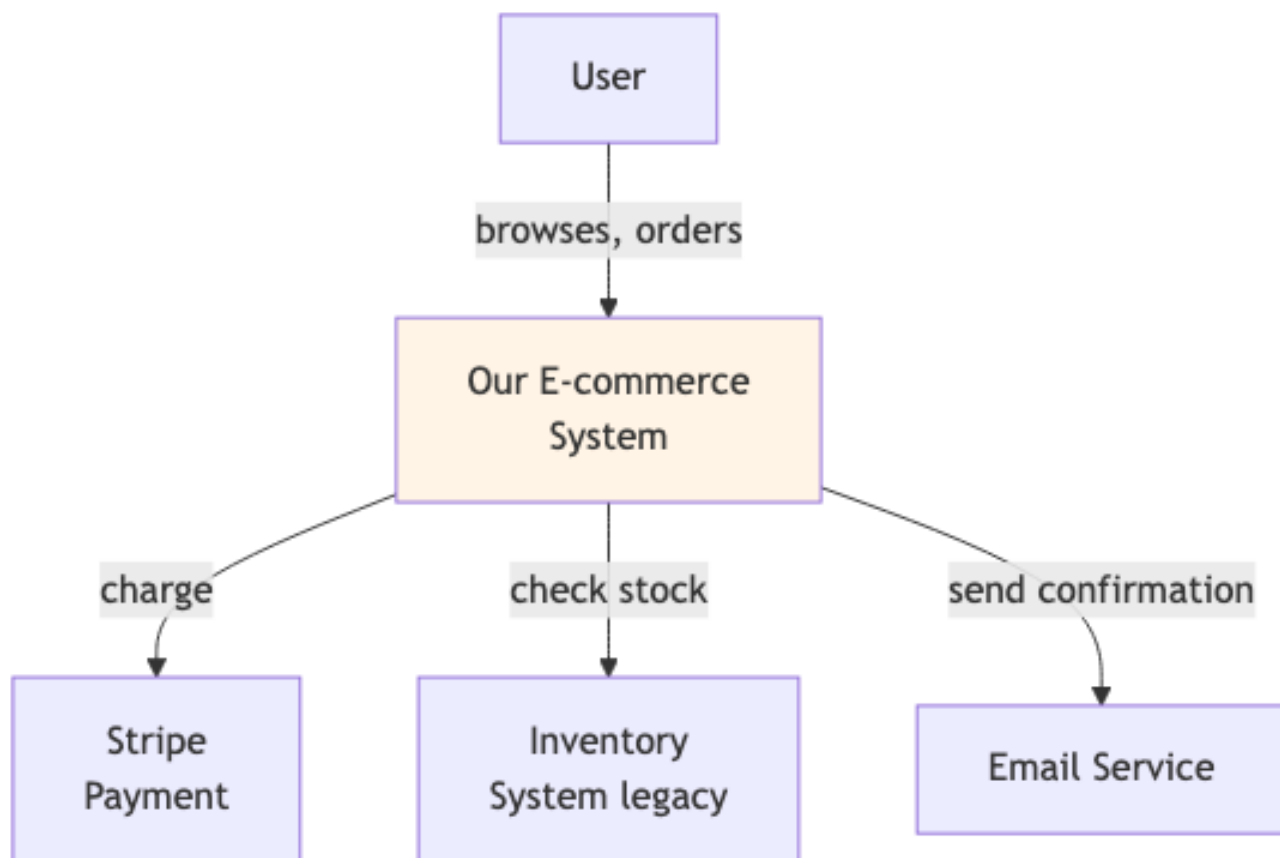
Audience: developer. Mục đích: hiểu code structure trong container.

Level 4: Code (optional)

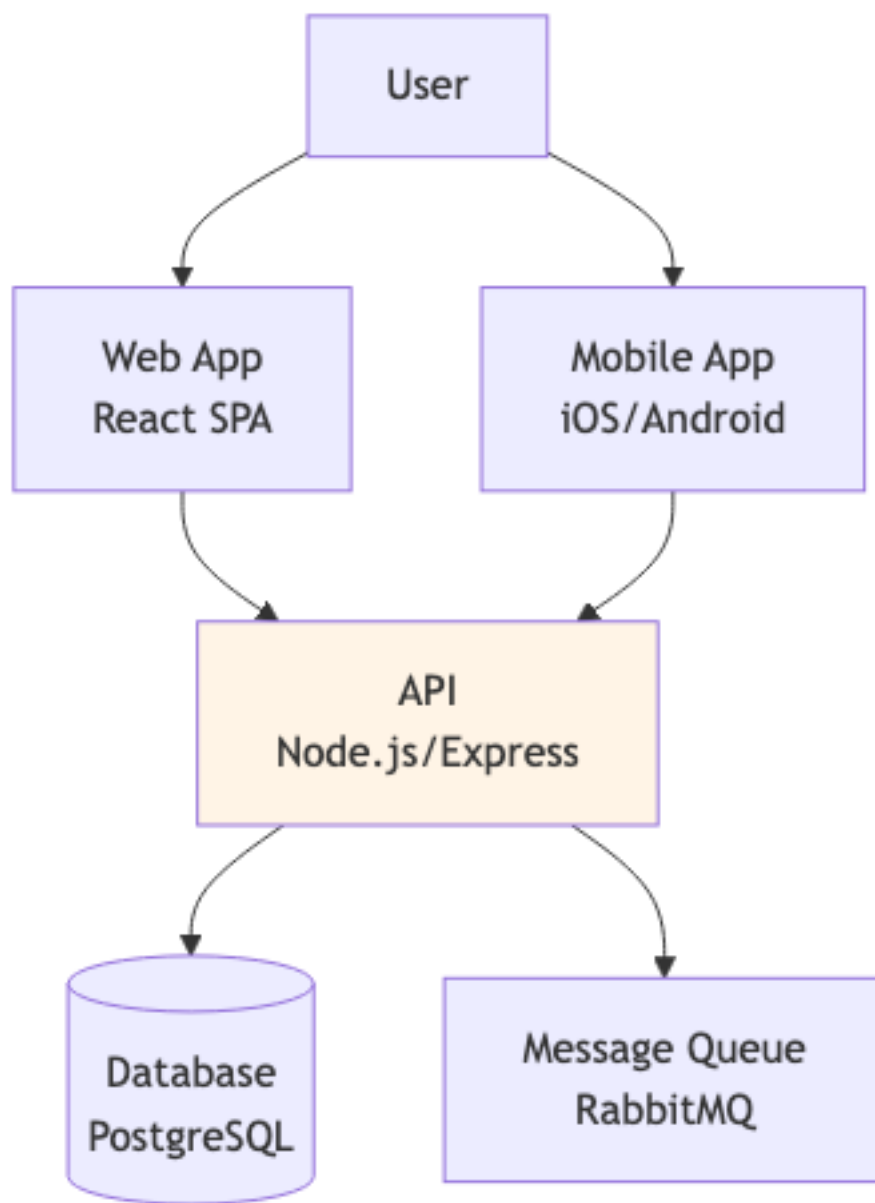
UML class diagram. Hiếm khi vẽ, code là source of truth.

Tools cho C4

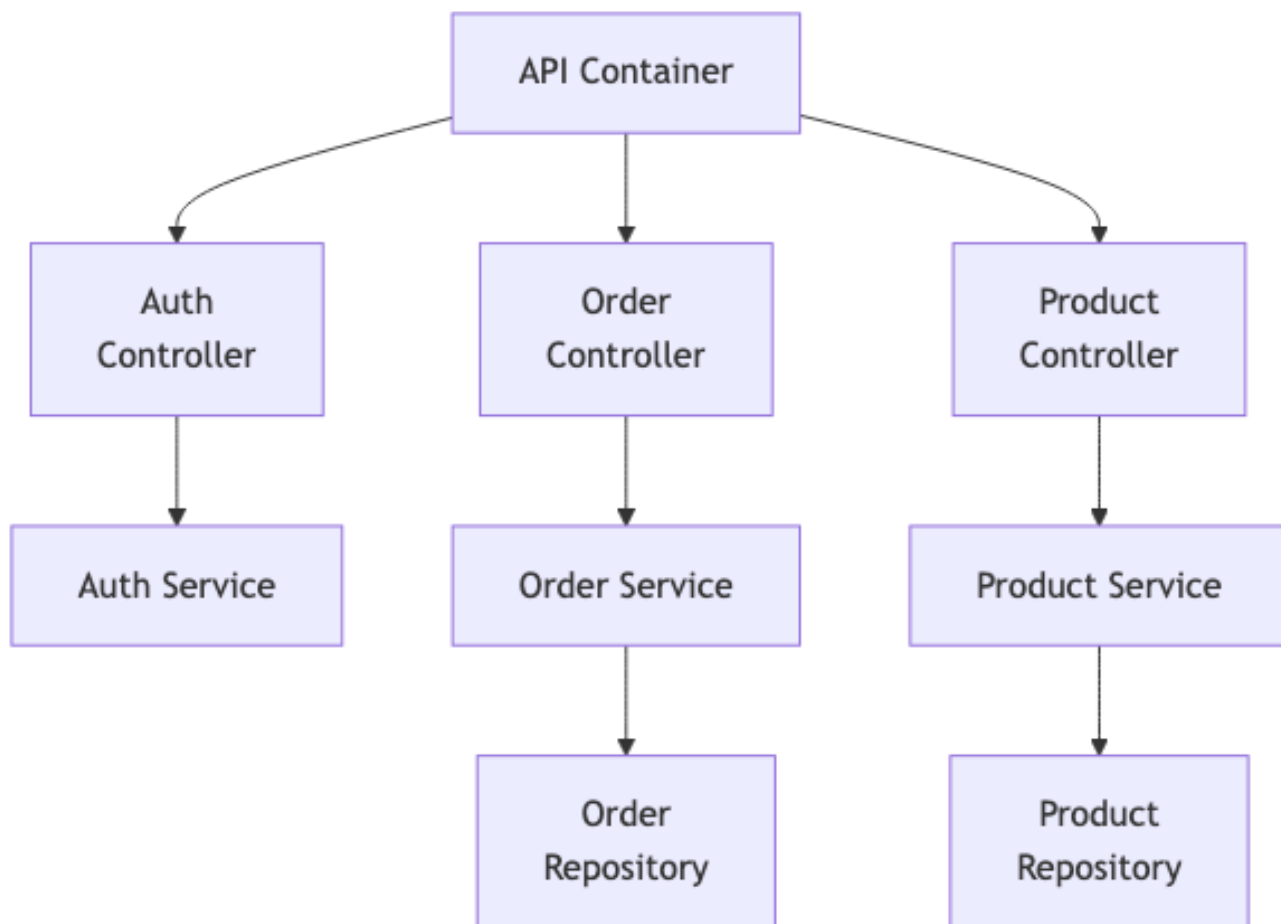
- **Structurizr DSL** (Simon Brown): text-based, render mọi level.
- **PlantUML + c4-plantuml**: include macro để vẽ C4.
- **Mermaid**: như example trên, để render trong markdown.



Hình 5: Sơ đồ 37



Hình 6: Sơ đồ 38



Hình 7: Sơ đồ 39

- **Draw.io**: GUI, nhưng không version-control friendly.

Recommendation: Structurizr DSL hoặc Mermaid + PlantUML.

ADR, Architecture Decision Record

ADR doc dạng:

```
# ADR-007: Use PostgreSQL for primary data store
```

```
## Status
```

Accepted (2026-01-15)

```
## Context
```

We need to select a primary database for the e-commerce platform. Expected scale:

- 50k users in year 1, target 500k by year 3.
- Complex queries (joins for product search, analytics).
- ACID required for orders and payments.
- Team experience: strong in SQL, basic in NoSQL.

```
## Decision
```

Use PostgreSQL 16 as primary data store. Use Redis for caching layer.

```
## Alternatives Considered
```

```
### MongoDB
```

- Pros: easier schema evolution, native JSON.
- Cons: weak ACID for cross-doc transactions, team less experienced.
- Rejected: ACID critical for payments; team expertise.

```
### MySQL
```

- Pros: similar SQL, mature.
- Cons: weaker JSON support, less feature-rich than Postgres.
- Rejected: Postgres has better JSONB, better extensions.

```
### DynamoDB
```

- Pros: managed, scalable.
- Cons: complex query model, vendor lock.
- Rejected: requires schema redesign around access patterns.

```
## Consequences
```

```
### Positive
```

- Strong ACID for orders/payments.
- Rich query support for analytics.
- Team productive immediately.

```
### Negative
```

- Scaling beyond ~1M users may require sharding (vs DynamoDB auto-scale).
- Operations overhead (backup, replication, tuning).

Mitigation

- Use managed Postgres (AWS RDS or Aurora) to reduce ops burden.
- Plan for read replica at 100k users.
- Plan for sharding strategy at 500k users.

ADR best practices

1. **Number sequentially**: ADR-001, ADR-002, ... never reuse numbers.
2. **Append-only**: Don't edit accepted ADRs. Supersede with new ADR.
3. **Short**: ≤ 1 page. Engineers won't read longer.
4. **Decision-focused**: 1 ADR = 1 decision, not analysis paper.
5. **Linked from code**: vd: PR mention ADR-007.
6. **Searchable**: keep in repo docs/adr/, in markdown.

Tools

- **adr-tools** (CLI): `adr new "Use PostgreSQL"`.
- **GitHub markdown**: render natively.
- **MADR template** (markdown ADR): widely-used template.

Practical: Documenting một service

Mức minimum cho 1 microservice:

1. **README.md**: setup, build, contribute.
2. **docs/architecture.md**: 1 page với:
 - System Context (where this service fits).
 - Container (deployment).
 - Key data models (top 5 entities).
 - Key flows (top 3-5 sequence).
3. **docs/adr/**: ADR file cho important decisions.
4. **API spec**: OpenAPI/protobuf in repo.
5. **runbook.md** (cho on-call): common ops procedures.

Tổng < 2 hours setup, value cho team rất cao.

Anti-patterns

Anti-pattern 1: Big upfront design

Write 200-page architecture document, then build. By time build done, doc outdated. No one reads.

Fix: living doc. Update with code changes. Use ADR for decisions.

Anti-pattern 2: Diagram-only doc

Doc chỉ diagram, không có narrative. Audience không hiểu context.

Fix: combine diagram + paragraph. "Đây là hệ X, làm Y, ở high level chia thành Z. [diagram]. Chi tiết: ...".

Anti-pattern 3: No audience awareness

Same doc cho mọi audience. Quá technical cho stakeholder, quá high-level cho dev.

Fix: layered doc. Stakeholder doc (C4 Context), Architect doc (C4 Container + ADR), Dev doc (C4 Component + code).

Anti-pattern 4: Doc rot

Doc viết 1 lần rồi không update. Sau 1 năm, doc $\neq$ code. Worse than no doc.

Fix: doc trong git. Code review verify doc updated. CI link checker.

Tóm tắt

- **Module Views**: static code structure. 4 subtypes.
- **C4 Model**: 4 mức zoom (Context-Container-Component-Code).
- **ADR**: 1-page decision record, append-only, in git.
- **Practical minimum**: README + architecture.md + ADR + API spec + runbook.
- **Tránh**: big upfront, diagram-only, no audience awareness, doc rot.

Bài tiếp: [C&C Views](#), runtime structure.

7.3 Component-and-Connector Views

Tóm tắt một dòng: C&C View show *runtime* structure - khi hệ chạy, component instances nào tồn tại, communicate với connector nào, data flow ra sao. Khác Module View ở chỗ Module = static, C&C = dynamic.

Nếu bạn đến từ Data Science

C&C View rất hữu ích để vẽ online inference. Một request đi từ client vào API, API gọi feature store, load model runtime, trả prediction, ghi prediction log, đẩy metric sang monitoring. Sequence này chính là nơi latency, failure và observability xuất hiện.

Nếu chỉ nhìn code training, bạn sẽ bỏ lỡ runtime behavior. C&C View giúp bạn thấy model không chạy một mình. Nó nằm trong chuỗi connector: HTTP, database query, cache lookup, message queue, logging pipeline.

Module vs C&C

Aspect	Module View	C&C View
Element	Module (class, package)	Component instance, connector
Relation	Depend, contain, inherit	Communicate at runtime
Time	Static (code)	Dynamic (runtime)
Example	Order package depends Payment package	OrderService instance calls PaymentService via HTTP
Tool	C4 Component diagram, UML class	Sequence diagram, runtime topology

Cùng hệ có thể có Module và C&C view *khác nhau*. Vd:

- Module: 1 payment module exist.
- C&C: 3 instance PaymentService chạy parallel (replicated for HA).

Subtypes của C&C

Subtype 1: Process View

Show processes/threads chạy ở runtime + sync/communication.

Purpose: hiểu request flow, latency contributors.

Subtype 2: Concurrency View

Show concurrent execution + synchronization.

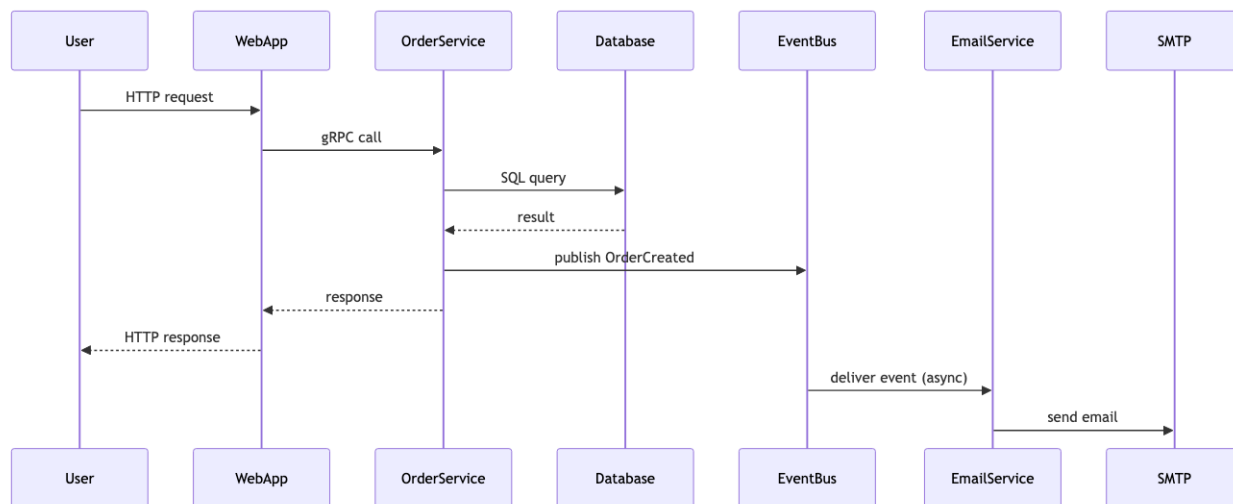
Vd:

Purpose: identify race condition, deadlock risk, sync points.

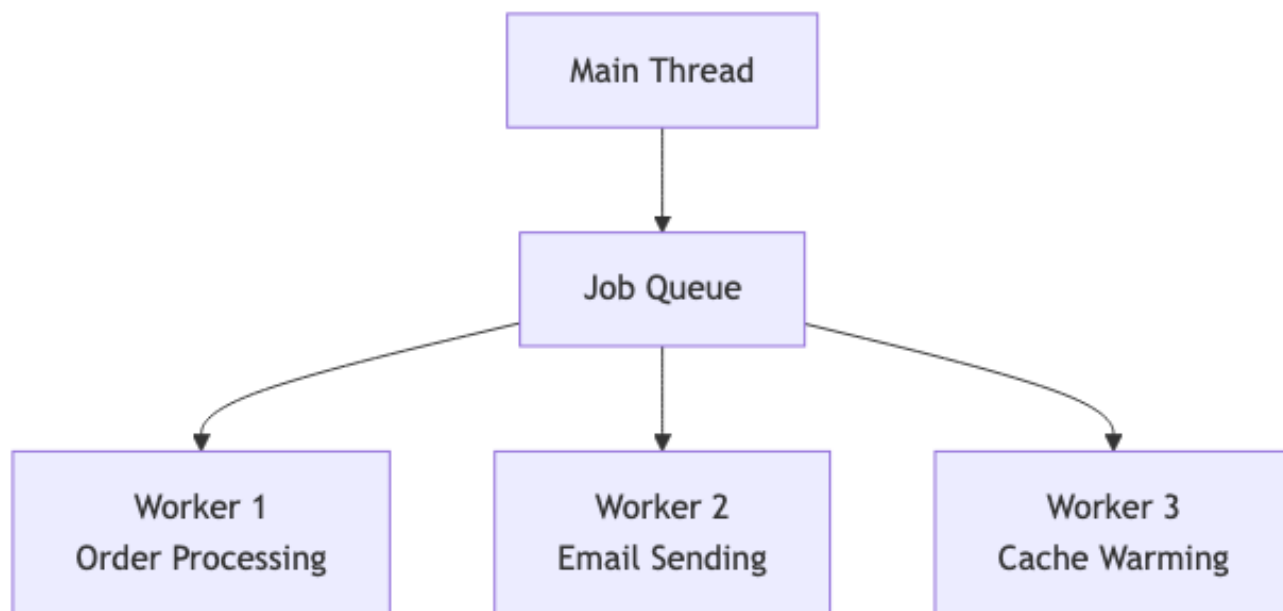
Subtype 3: Communication View

Show topology của component instances + communication links.

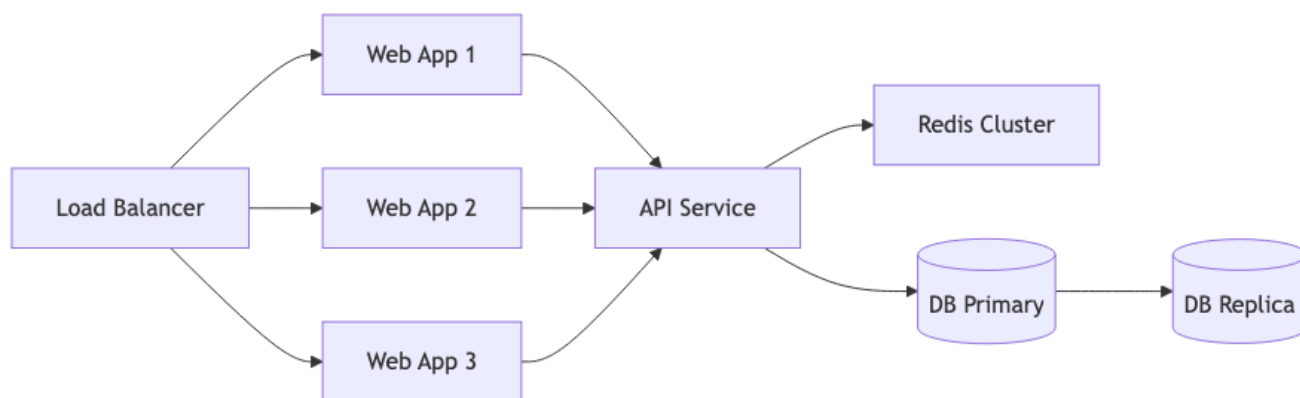
Purpose: visualize HA, scalability, single points of failure.



Hình 8: Sơ đồ 40



Hình 9: Sơ đồ 41



Hình 10: Sơ đồ 42

Subtype 4: Deployment-time

Same as Allocation View (Bài 7.4). Where component instances run physically.

Khi nào dùng C&C View

- **Debug performance:** trace request flow.
- **Identify bottleneck:** where data flows slow.
- **Design HA:** show replication, failover paths.
- **Onboard SRE:** where data flows through stack.
- **Capacity planning:** scale point identification.

Diagram types**Sequence Diagram (UML)**

Time vs participant. Show message exchange.

Pros: rất intuitive cho dynamic behavior.

Cons: only 1 scenario per diagram. Many scenarios = many diagrams.

Tools: Mermaid, PlantUML, draw.io.

Activity Diagram (UML)

Flow chart cho process. Decision, loop, parallel.

Pros: tốt cho business workflow.

Cons: less precise for technical detail.

Communication Diagram (UML)

Component + numbered messages.

Pros: emphasize structure.

Cons: less popular than sequence.

Runtime Topology Diagram

Boxes + connectors, custom notation.

Pros: flexible.

Cons: không standard, learning curve cho reader.

C4 Container diagram cũng là dạng C&C View ở mức container.

Connector types

Connector = mechanism component giao tiếp:

- **Method call** (in-process).
- **HTTP/REST**: sync, request/response.
- **gRPC**: sync, binary protocol.
- **Message queue**: async (Kafka, RabbitMQ).
- **Event bus**: async pub/sub (EDA).
- **Database call**: query.
- **File system**: read/write file.
- **Shared memory**: in-process or IPC.

C&C diagram nên *label* connector với type:

```
[OrderService] --HTTP--> [PaymentService]
```

```
[OrderService] --Kafka--> [EmailService]
```

```
[WebApp] --gRPC--> [API]
```

Label giúp reader hiểu communication semantics (sync vs async, latency tier).

Documenting common runtime patterns

Pattern 1: Request flow

Trace 1 user request qua stack:

Use case: explain to new dev, debug latency issue.

Pattern 2: Failover

Use case: explain HA strategy to architect review.

Pattern 3: Saga (distributed transaction)

Đã show ở Bài 6.4.

Pattern 4: Async event flow

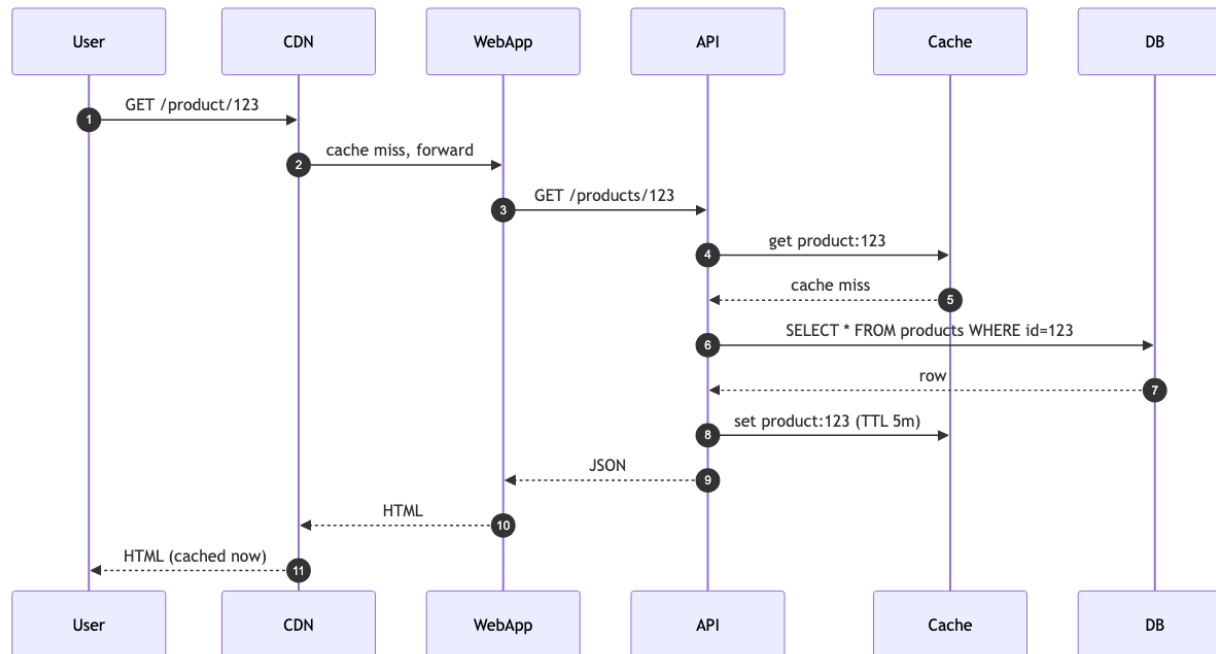
Use case: explain decoupling of producer from consumers.

Sai lầm thường gặp

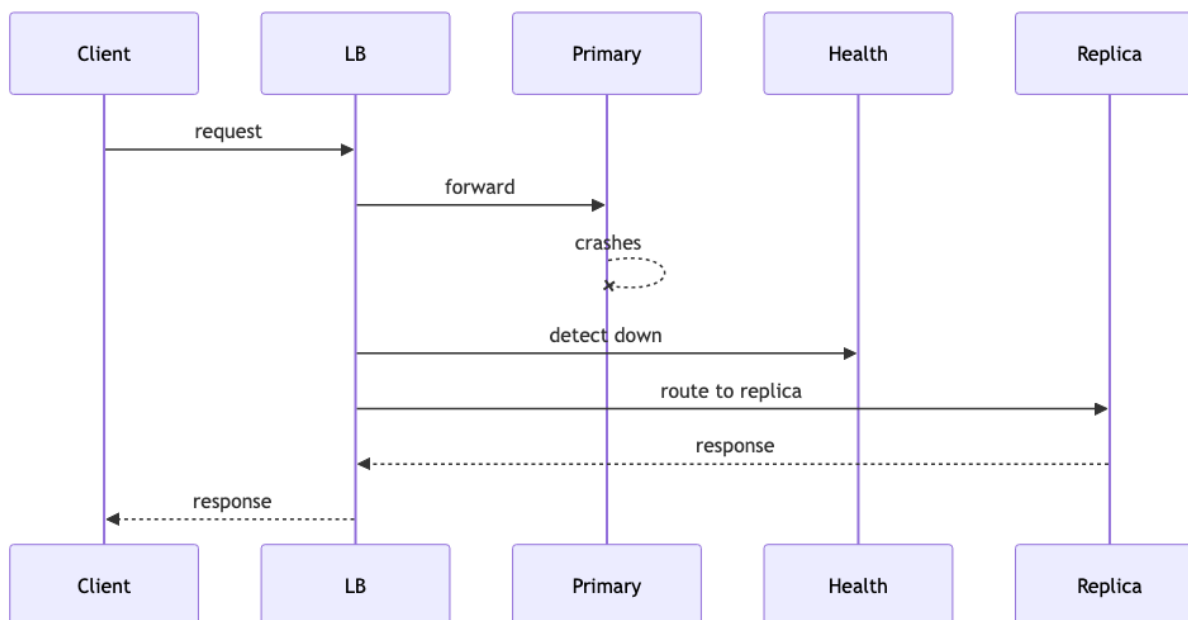
Sai lầm 1: Vẽ mọi sequence

100 sequence diagrams cho 100 endpoints. Maintain nightmare.

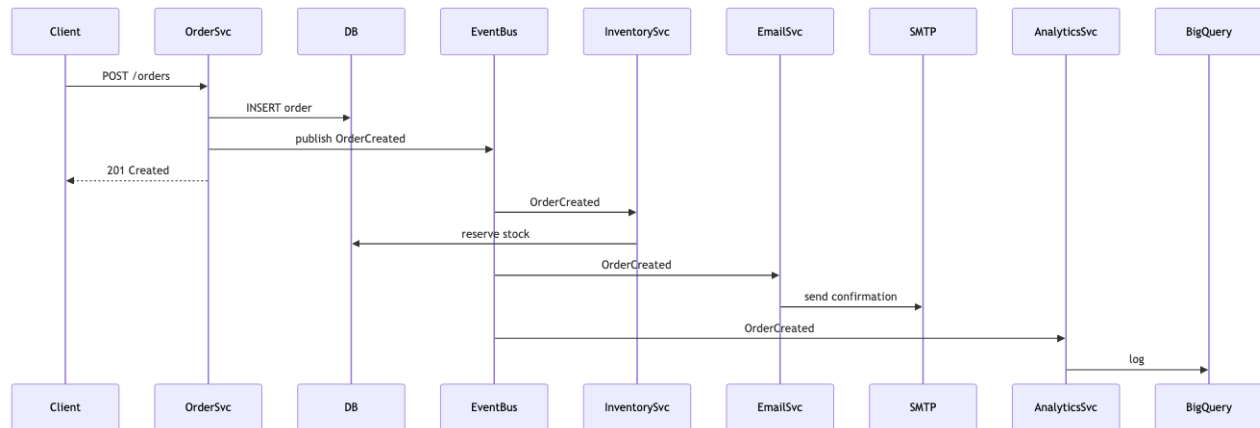
Fix: chỉ vẽ top 5-10 critical/complex flows. Đơn giản flows = doc trong API spec.



Hình 11: Sơ đồ 43



Hình 12: Sơ đồ 44



Hình 13: Sơ đồ 45

Sai lầm 2: Quên label connector

“A □ B” không biết HTTP hay async event. Reader misunderstand.

Fix: always label arrow type.

Sai lầm 3: Mixed level of abstraction

Một diagram show high-level (services) lẫn low-level (function call). Confusing.

Fix: each diagram one level. C4 model là nguyên tắc tốt.

Sai lầm 4: No legend

Diagram phức tạp với colors/shapes mà không có legend. Reader guess meaning.

Fix: legend mọi non-standard notation.

Sai lầm 5: Outdated runtime

Code thay đổi, diagram không update. Worse than no diagram.

Fix: include diagram update in PR template for major changes.

Tóm tắt

- **C&C View**: runtime structure (vs Module = static).
- **4 subtypes**: Process, Concurrency, Communication, Deployment.
- **Diagram types**: sequence (most popular), communication, runtime topology.
- **Always label connectors** với type (HTTP/Kafka/gRPC/...).
- **Tránh**: vẽ mọi sequence, mixed abstraction, outdated.

Bài tiếp: [Allocation Views](#), physical deployment.

7.4 Allocation Views

Tóm tắt một dòng: Allocation View show *physical* mapping của software lên: hardware/cloud (deployment), file system (implementation), team (work assignment). Trả lời “code này chạy ở đâu, ai own nó”.

Nếu bạn đến từ Data Science

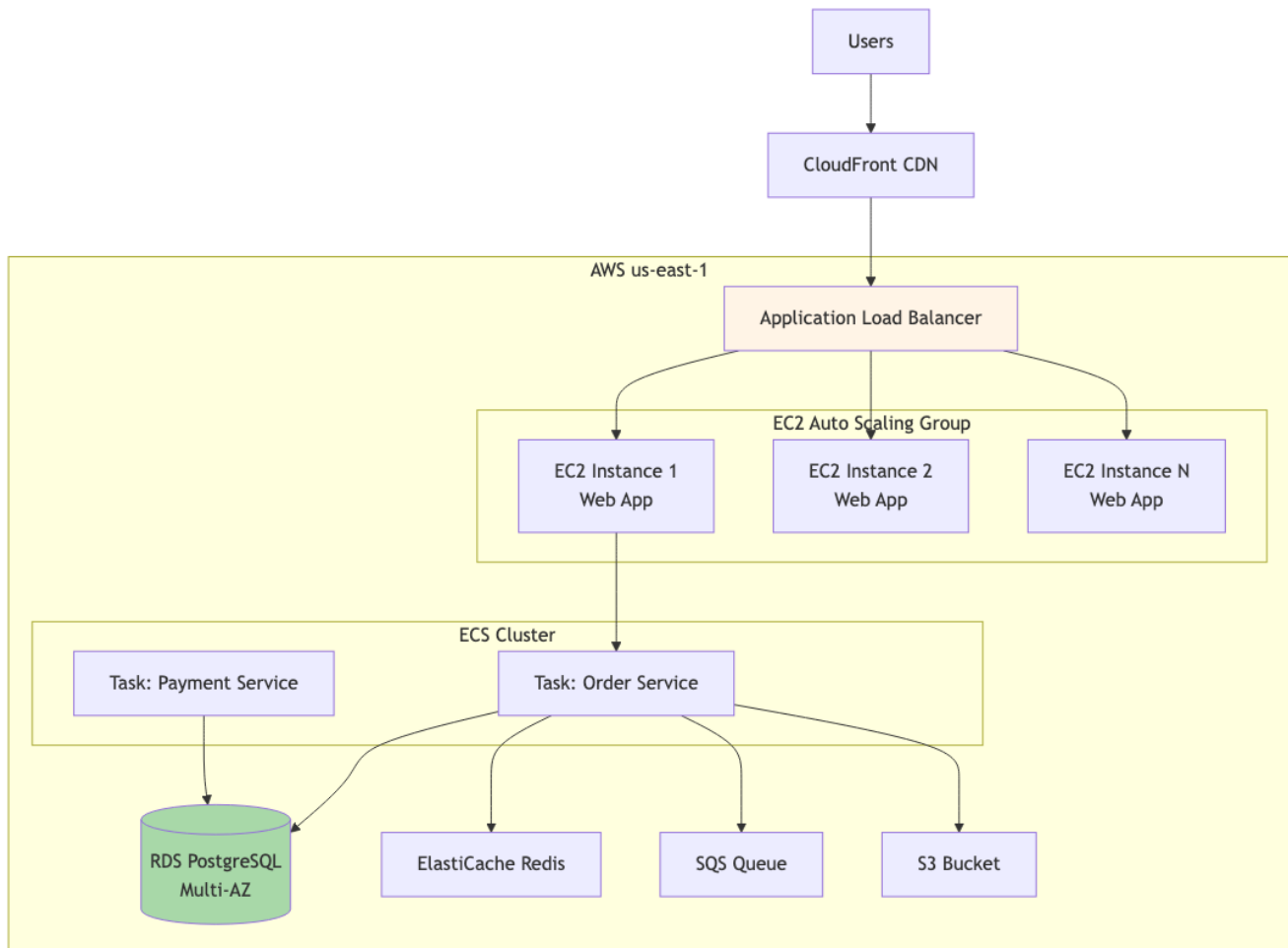
Allocation View trả lời câu hỏi model và data pipeline chạy ở đâu. Training job có chạy trên GPU node không? Serving API chạy CPU hay GPU? Feature store nằm trong Redis, online DB hay managed service? Data lake ở object storage nào? Monitoring chạy chung cluster hay riêng?

Những quyết định này ảnh hưởng cost, latency và recoverability. Một model tốt nhưng deploy nhầm lên instance quá yếu vẫn làm hệ thống fail.

3 subtypes

Subtype 1: Deployment View

Mapping software $\square$ hardware/cloud infrastructure.



Hình 14: Sơ đồ 46

Audience: SRE, DevOps, architect.

Purpose:

- Capacity planning.
- Cost estimation.
- Failure scenario analysis.
- Migration planning.

Subtype 2: Implementation View

Mapping software $\square$ file system / repository structure.

```
e-commerce-platform/
├─ apps/
│   ├─ web/                # React SPA
│   ├─ mobile/             # React Native
│   └─ admin/              # Admin UI
├─ services/
│   ├─ order-service/      # Go service
│   ├─ payment-service/    # Java service
│   ├─ inventory-service/  # Python service
│   └─ notification-service/ # Node.js service
├─ shared/
│   ├─ proto/              # gRPC schemas
│   ├─ events/             # Event schemas
│   └─ libs/               # Shared libraries
├─ infra/
│   ├─ terraform/          # Infrastructure as Code
│   ├─ helm/               # K8s charts
│   └─ monitoring/         # Grafana, Prometheus config
├─ docs/
│   ├─ architecture/       # ADRs, diagrams
│   └─ runbooks/           # Ops runbooks
└─ README.md
```

Audience: developer, new joiner.

Purpose:

- Navigation: “find code”.
- Mono vs multi repo decision documentation.

Subtype 3: Work Assignment View

Mapping software $\square$ team ownership.

Team	Owns
Order team (5 engineers)	order-service, order-events, order admin UI
Payment team (4 engineers)	payment-service, payment integrations
Inventory team (3 engineers)	inventory-service, warehouse interface
Platform team (6 engineers)	shared libs, monitoring, CI/CD

Team	Owns
Web team (4 engineers)	web SPA, mobile, design system

Audience: Engineering Manager, Director, new joiner.

Purpose:

- On-call rotation.
- Code review routing.
- Hiring plan.
- Conway's law analysis.

Cloud topology patterns

Common patterns trong deployment view:

Pattern 1: Single region, multi-AZ

Hệ chạy ở 1 region (us-east-1), span 3 availability zones cho HA.

Use case: hầu hết app medium scale, không cần multi-region.

Cost: medium. Failover: AZ-level (region down = total down).

Pattern 2: Multi-region active-passive

Primary region serves traffic. Secondary region standby, can take over within minutes.

Use case: critical app cần disaster recovery.

Cost: high (secondary region underutilized).

Pattern 3: Multi-region active-active

Both regions serve traffic. Replication bidirectional.

Use case: global app, low latency cho global users.

Cost: highest. Complexity: high (data sync).

Pattern 4: Multi-cloud

Spread across AWS, GCP, Azure.

Use case: avoid vendor lock-in, regulatory.

Cost: very high. Operational complexity: very high.

Pattern 5: Edge + Cloud

Static assets ở CDN edge (CloudFront, Fastly). Dynamic logic ở core cloud.

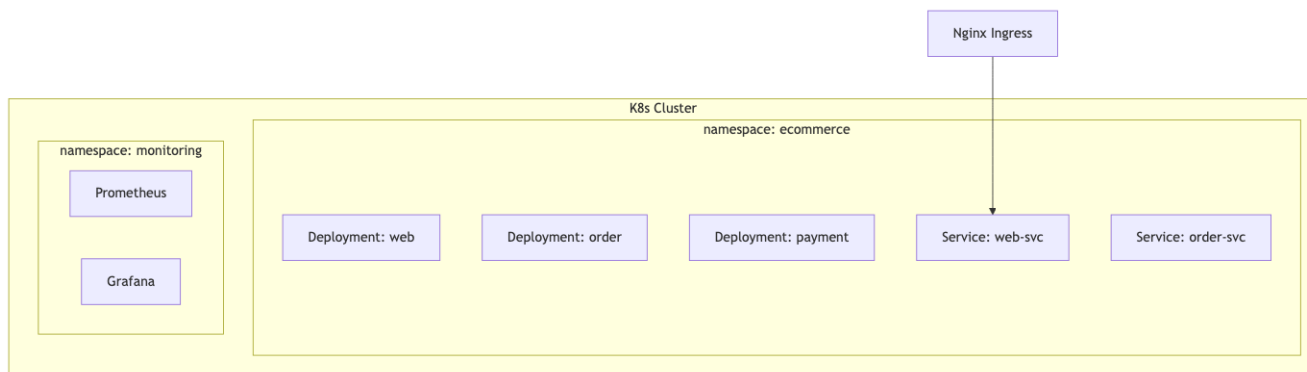
Use case: any web app. Standard practice.

Container orchestration

Modern deployment thường dùng container orchestration:

Kubernetes (K8s)

Standard cho large-scale container deployment. Show as:



Hình 15: Sơ đồ 47

Audience: SRE, K8s admin.

Managed services

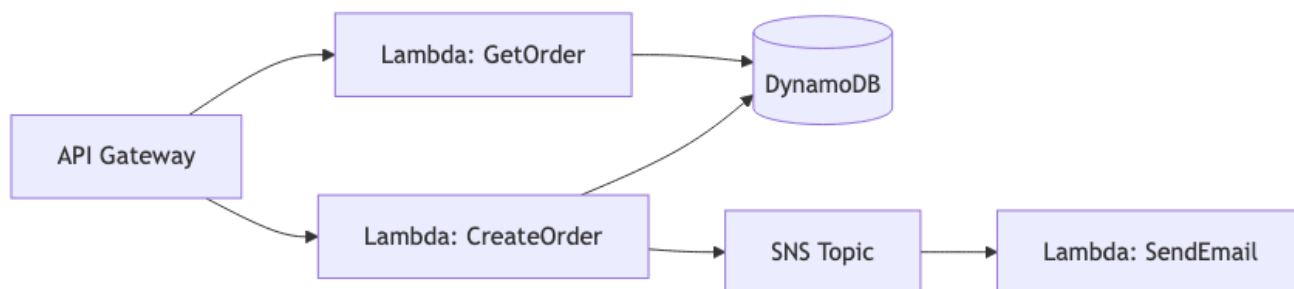
Cloud-native: AWS App Runner, GCP Cloud Run, Fly.io. Simpler ops, less control.

Use khi: team không có K8s expertise.

Serverless

AWS Lambda, GCP Cloud Functions. No server management.

Show as:



Hình 16: Sơ đồ 48

Use case: event-driven, low-medium traffic, cost-sensitive.

Conway's Law

Melvin Conway (1968):

“Organizations which design systems are constrained to produce designs which are copies of the communication structures of these organizations.”

Diễn đạt: architecture của hệ phản ánh org chart của team build.

Examples

- 3-team org (frontend, backend, mobile) □ 3-tier architecture với boundary tại tech layer.
- Bounded context per team □ microservices boundary follow team.
- Centralized DBA team □ shared database (DBA gatekeeper).

Implications

- **Architecture decision = org decision.** Đổi từ monolith □ microservices ≈ đổi org từ functional □ domain team.
- **Inverse Conway maneuver:** tổ chức org theo mong muốn architecture, để Conway force tự nhiên tạo ra architecture đó.

Visualization

Work Assignment View chính là Conway's law trong tài liệu. Show team-component mapping □ architect predict architecture evolution.

Documenting cloud cost

Allocation View nên include cost (estimate):

Component	Instance	Cost/month
Web App (3x EC2 t3.medium)	EC2	\$90
Order Service (ECS Fargate, 2 task)	ECS	\$120
Database (RDS PostgreSQL, m5.large)	RDS	\$250
Redis (ElastiCache cache.t3.small)	ElastiCache	\$25
Load Balancer	ALB	\$20
Data transfer (1TB out)	Network	\$90
Total		~\$595/month

Cost transparency helps:

- Architect cân nhắc cost trong decision (Bài 3.3 trade-off).
- Engineering Manager budget planning.
- Identify waste (under-utilized resources).

Tools: AWS Cost Explorer, Cloud Custodian, custom dashboard.

Sai lầm thường gặp

Sai lầm 1: Outdated deployment diagram

Infra thay đổi (added Redis, swapped DB), diagram không update.

Fix: infra-as-code (Terraform) is source of truth. Diagram auto-generate from IaC nếu possible.

Sai lầm 2: No cost annotation

Diagram show topology mà không cost. Decision purely technical, ignore \$\$.

Fix: annotate cost per component (rough estimate OK).

Sai lầm 3: Quên failure scenario

Diagram show happy path. Không show “DB down thì sao”, “AZ down thì sao”.

Fix: separate diagram cho failure scenarios. Annotate SLA.

Sai lầm 4: Work Assignment không up-to-date

Org chart thay đổi (team merge, split), document outdated.

Fix: review work assignment view quarterly. Update on org change.

Sai lầm 5: Quên Conway’s law

Đổi architecture mà không đổi team. Distributed monolith result.

Fix: align team boundary với architecture boundary. Inverse Conway.

Tóm tắt

- **Allocation Views:** software $\square$ physical/team mapping.
- **3 subtypes:** Deployment (hardware), Implementation (file system), Work Assignment (team).
- **Cloud patterns:** single/multi region, active-active, multi-cloud, edge+cloud.
- **Conway’s law:** architecture follows org chart. Inverse Conway = design org for desired architecture.
- **Include cost** trong deployment view.

Tổng kết Phần 7

3 view chính từ SEI:

View	Câu hỏi	Audience
Module	Code organize ra sao? Depend ra sao?	Developer
C&C	Runtime ra sao? Communicate ra sao?	Architect, SRE
Allocation	Chạy ở đâu? Ai own?	DevOps, EM, Stakeholder

Plus ADR cho decision history.

Practical minimum: System Context (C4) + Container (C4) + 2-3 sequence diagrams cho critical flows + ADR cho important decisions.

Phần tiếp: [Phần 8 - Case Studies](#), apply tất cả vào case thực.